House Price Analysis

Project 6 — Data Analysis Portfolio

Executive Summary

This project analyzes 300 residential properties to understand the factors associated with property prices. The analysis covers property size, bedrooms, bathrooms, age, location, property type, price segments, and price per area. The strongest numerical relationship with price is property area, while location shows substantial differences in average prices.

Key Performance Indicators

Metric	Result
Properties analyzed	300
Average price	■2.49 crore
Median price	■2.24 crore
Minimum price	■36.95 lakh
Maximum price	■5.87 crore
Area–Price correlation	0.796
Average price per area	■9,542.84

Methodology

The dataset was loaded and validated using pandas. Descriptive statistics, correlation analysis, grouped aggregations, price segmentation, and price-per-area analysis were performed. Seven visualizations were created using Matplotlib to communicate the findings.

Major Findings

• Property area has a strong positive correlation with price (0.796).

• City Center properties have the highest average price at approximately ■3.31 crore.

• Suburban properties average approximately ■2.51 crore, while Rural properties average approximately ■1.65 crore.

• Apartments have the highest average price among property types at approximately ■2.71 crore.

• Premium properties average approximately ■4.07 crore and have much larger average areas than Budget properties.

• New properties have higher average prices than older properties, but age is a weaker factor than area.

• The lowest price-per-area observations are concentrated in Rural locations.

Location and Property Type Analysis

Location	Count	Average Price	Average Area
City Center	97	■3.31 crore	2,777.87
Suburb	105	■2.51 crore	2,764.89

Rural	98	■1.65 crore	2,736.16

Business Recommendations

• Use area and location as primary inputs when estimating asking prices.

• Compare properties using price per area as well as total price.

• Target Rural and Suburban properties for buyers prioritizing space and affordability.

• Use Budget, Mid-Range, and Premium segments for targeted marketing.

• Treat property age as a secondary pricing factor rather than a standalone predictor.

• Build dashboards tracking price, area, location, property type, age, and price per area.

Data Quality and Validation

Validation identified 300 records, 0 missing values, 0 duplicate rows, and 300 unique Property IDs. The original numerical and categorical fields were validated before derived features were created.

Visualizations

• House Price Distribution

• Area vs House Price

• Average House Price by Location

• Average House Price by Property Type

• Average Price by Price Segment

• Average House Price by Number of Bedrooms

• Average House Price by Property Age Group

Conclusion

The analysis shows that property area and location are the strongest practical pricing signals in this dataset. Combining these factors with price segmentation and price-per-area analysis provides useful perspectives for buyers, sellers, and real estate businesses.

Note: This dataset is used for analytical and portfolio purposes. The identified relationships represent associations within the dataset and should not be interpreted as causal relationships.